

HDOC – Experiment 6

Question 1

Write a C program to find linear velocity when radius and revolutions per second are given.

❖ Understanding problem

Input

- Radius (r) in meters
- Revolutions per second (n)

Output

- Linear Velocity (m/s)

Formula

Angular Velocity: $\omega = 2\pi n$

Linear velocity: $v = r\omega = 2\pi rn$

❖ Test case table

Test Case	Input (radius, RPS)	Expected output	Actual output	Result
1	r = 2 m n = 5	62.831 m/s	Same	Pass
2	r = 1.5 m n = 10	94.247 m/s	Same	Pass
3	r = 0.5 m n = 8	25.132 m/s	Same	Pass

❖ Algorithm

1. Start.
2. Declare variables for radius, revolutions per second, and linear velocity.
3. Read the radius and revolutions per second.
4. Calculate linear velocity using:
$$v = 2\pi rn$$
5. Display the linear velocity.
6. Stop.

❖ Program code in C

```
#include <stdio.h>

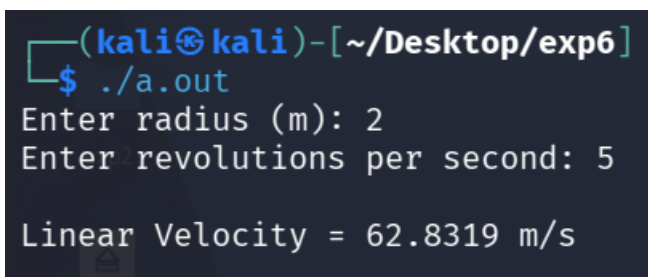
#define PI 3.141592653589793

int main()
{
    double radius, rps, velocity;
```

```
printf("Enter radius (m): ");  
scanf("%lf", &radius);  
  
printf("Enter revolutions per second: ");  
scanf("%lf", &rps);  
  
velocity = 2 * PI * radius * rps;  
  
printf("\nLinear Velocity = %.4lf m/s\n", velocity);  
  
return 0;  
}
```

❖ Compile, Execute and Evaluate

Test case 1



```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter radius (m): 2  
Enter revolutions per second: 5  
  
Linear Velocity = 62.8319 m/s
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter radius (m): 1.5  
Enter revolutions per second: 10  
  
Linear Velocity = 94.2478 m/s
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter radius (m): 0.5  
Enter revolutions per second: 8  
  
Linear Velocity = 25.1327 m/s
```

EVALUATION: All the three cases produces the expected results, therefore, the program passes all the cases.

❖ Compilation and Execution screenshot

```
(kali㉿kali)-[~/Desktop/exp6]
$ vi velocity.c

(kali㉿kali)-[~/Desktop/exp6]
$ cc velocity.c

(kali㉿kali)-[~/Desktop/exp6]
$ ./a.out
Enter radius (m): 2
Enter revolutions per second: 5

Linear Velocity = 62.8319 m/s

(kali㉿kali)-[~/Desktop/exp6]
$ ./a.out
Enter radius (m): 1.5
Enter revolutions per second: 10

Linear Velocity = 94.2478 m/s

(kali㉿kali)-[~/Desktop/exp6]
$ ./a.out
Enter radius (m): 0.5
Enter revolutions per second: 8

Linear Velocity = 25.1327 m/s
```

Question 2

Write a C program to find electric power using voltage and resistance.

❖ Understanding problem

Input

- Voltage (V) in volts
- Resistance (R) in ohms

Output

- Electric Power (P) in watts

Formula

$$P = V^2/R$$

❖ Test case table

Test Case	Input (Voltage, Resistance)	Expected output	Actual output	Result
1	220 V, 110Ω	440W	440W	Pass
2	120V, 60Ω	240W	240W	Pass
3	100V, 25Ω	400W	400W	Pass

❖ Algorithm

1. Start.
2. Declare voltage, resistance and power.
3. Read voltage and resistance.
4. Calculate power using:

$$P = V^2/R$$

5. Display the power.
6. Stop.

❖ Program code in C

```
#include <stdio.h>

int main()
{
    double voltage, resistance, power;

    printf("Enter voltage (V): ");
    scanf("%lf", &voltage);

    printf("Enter resistance (Ohms): ");
    scanf("%lf", &resistance);

    if (resistance == 0)
    {
        printf("Resistance cannot be zero.\n");
        return 1;
    }
}
```

```
}

power = (voltage * voltage) / resistance;

printf("\nElectric Power = %.2lf Watts\n", power);

return 0;
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp6]
$ ./a.out
Enter voltage (V): 220
Enter resistance (Ohms): 110

Electric Power = 440.00 Watts
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp6]
$ ./a.out
Enter voltage (V): 120
Enter resistance (Ohms): 60

Electric Power = 240.00 Watts
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter voltage (V): 100  
Enter resistance (Ohms): 25  
  
Electric Power = 400.00 Watts
```

EVALUATION: All the three cases produces the expected results, therefore, the program passes all the cases.

❖ Compilation and Execution screenshot

```
(kali㉿kali)-[~/Desktop/exp6]  
$ vi power.c  
  
(kali㉿kali)-[~/Desktop/exp6]  
$ cc power.c  
  
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter voltage (V): 220  
Enter resistance (Ohms): 110  
  
Electric Power = 440.00 Watts  
  
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter voltage (V): 120  
Enter resistance (Ohms): 60  
  
Electric Power = 240.00 Watts  
  
(kali㉿kali)-[~/Desktop/exp6]  
$ ./a.out  
Enter voltage (V): 100  
Enter resistance (Ohms): 25  
  
Electric Power = 400.00 Watts
```

Question 3

Write a C program to find the resultant magnitude of two perpendicular vectors.

❖ Understanding problem

Input

- Magnitude of first vector
- Magnitude of second vector

Output

- Resultant Magnitude

Formula

$$R = \sqrt{A^2 + B^2}$$

❖ Test case table

Test Case	Input (A,B)	Expected output	Actual output	Result
1	3, 4	5.0000	Same	Pass
2	5, 12	13.0000	Same	Pass
3	8, 15	17.0000	Same	Pass

❖ Algorithm

1. Start.
2. Declare variables for the two vectors and resultant.
3. Read both vector magnitudes.
4. Calculate:

$$R = \sqrt{A^2 + B^2}$$

5. Display the resultant magnitude.
6. Stop.

❖ Program code in C

```
#include <stdio.h>

#include <math.h>

int main()
{
    double vector1, vector2, resultant;

    printf("Enter first vector: ");
    scanf("%lf", &vector1);

    printf("Enter second vector: ");
    scanf("%lf", &vector2);

    resultant = sqrt((vector1 * vector1) + (vector2 * vector2));

    printf("\nResultant Magnitude = %.4lf\n", resultant);
```

```
    return 0;  
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./vectors  
Enter first vector: 3  
Enter second vector: 4  
  
Resultant Magnitude = 5.0000
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./vectors  
Enter first vector: 5  
Enter second vector: 12  
  
Resultant Magnitude = 13.0000
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp6]  
$ ./vectors  
Enter first vector: 8  
Enter second vector: 15  
  
Resultant Magnitude = 17.0000
```

EVALUATION: All the three cases produces the expected results, therefore, the program passes all the cases.

❖ Compilation and Execution screenshot

```
(kali㉿kali)-[~/Desktop/exp6]
$ vi vectors.c

(kali㉿kali)-[~/Desktop/exp6]
$ gcc vectors.c -o vectors -lm

(kali㉿kali)-[~/Desktop/exp6]
$ ./vectors
Enter first vector: 3
Enter second vector: 4

Resultant Magnitude = 5.0000

(kali㉿kali)-[~/Desktop/exp6]
$ ./vectors
Enter first vector: 5
Enter second vector: 12

Resultant Magnitude = 13.0000

(kali㉿kali)-[~/Desktop/exp6]
$ ./vectors
Enter first vector: 8
Enter second vector: 15

Resultant Magnitude = 17.0000
```